

Topology by Marco Manetti, Exercises

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1 Geometrical Introduction

Suppose Γ is a connected graph where each node has even degree. Prove that every graph obtained by removing one edge from Γ is still connected.

Proof. To show that Γ is still connected, we have to show that there exists a walk from v_0 to v_k where v_0, v_k are the nodes between which an edge has been removed. Since before the removal we have an Eulerian graph, we consider the edge that is removed to be the last edge in the walk. We still have a walk that reaches every point in the graph, since every edge has been traversed. $\square$